Hyperuniformity in Chemotactic Active Matter

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Contents

1 Models

1.1 Particle-based model

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D}\boldsymbol{\xi}(t) \ , \tag{1a}$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + k \sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t)) \ , \tag{1b}$$

for $i = 1, 2, \cdots, N$. Here, $\mathbf{r}_i$ is the position of the i-th particle, c is the concentration field of the chemoattractant, α is the chemotactic sensitivity. If $\alpha > 0$, the particles are attracted to higher concentrations of the chemoattractant (chemoattraction/positive chemotaxis), while if $\alpha < 0$, the particles are repelled by higher concentrations of the chemoattractant (chemorepulsion/negative chemotaxis). D is the diffusion coefficient of the particles, $\boldsymbol{\xi}$ is a Gaussian white noise with zero mean and unit variance, D_c is the diffusion coefficient of the chemoattractant, μ is the decay ($\mu < 0$) or growth ($\mu > 0$) rate of the chemoattractant, and k is the production ($k > 0$) or consumption rate ($k < 0$) of the chemoattractant by the particles.

For simplicity, we assume that particles are point-like and move in a two-dimensional $L \times L$ square domain with periodic boundary conditions. The initial positions of the particles can be randomly distributed, and the initial concentration field can be set to a small random perturbation around a uniform background concentration c_0.

To reduce the parameter space, we can non-dimensionalize the equations by introducing $x_0 = L$, $t_0 = 1/k$ and c_0 as characteristic scales of length, time and concentration. The resulting dimensionless parameters are $\tilde{\alpha} = \frac{\alpha c_0}{L^2 k}$, $\tilde{D} = \frac{D}{L^2 k}$, $\tilde{D}_c = \frac{D_c}{L^2 k}$, and $\tilde{\mu} = \frac{\mu}{k}$. We can omit the tildes and work with the dimensionless equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}_i(t) = \alpha \nabla c(\mathbf{r}_i(t), t) + \sqrt{2D}\boldsymbol{\xi}(t) \ , \tag{2a}$$

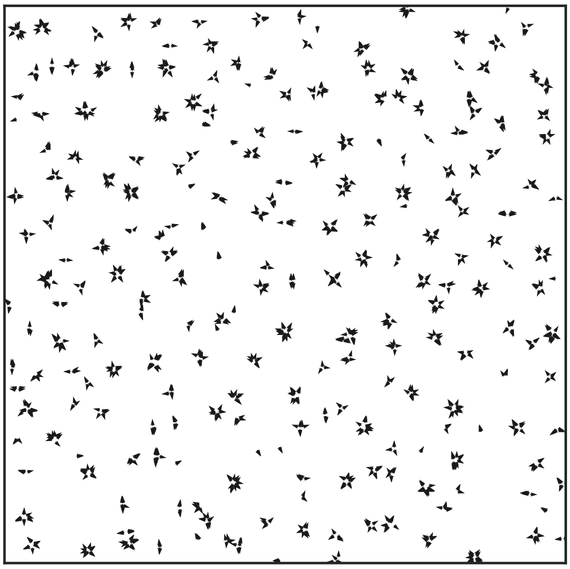
$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) + \sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t)) \ . \tag{2b}$$

Note that the above equations mathematically implies the assumption of $k > 0$. If $k < 0$, the non-dimensionalization would be different, and the Eq. (2b) would have a negative sign in front of the summation term:

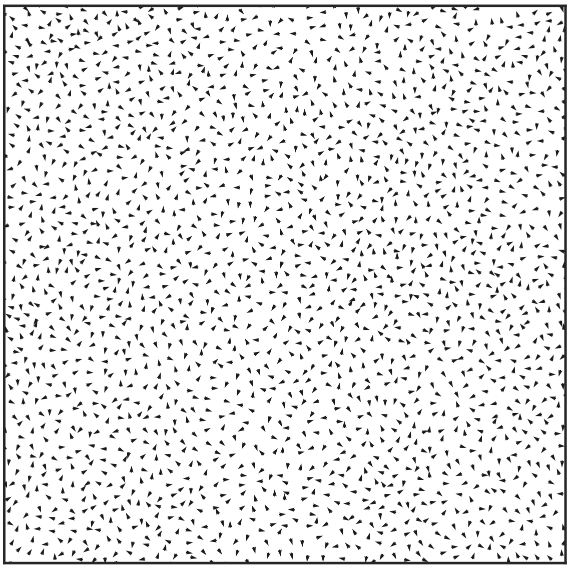
$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c(\mathbf{r}, t) + \mu c(\mathbf{r}, t) - \sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t)) \ . \tag{3}$$

In the following, we will focus on the case of $k < 0$ (chemorepulsion/negative chemotaxis), which is more relevant to the emergence of hyperuniformity.

As shown in the snapshots, the case of production ($k > 0$) leads to aggregation of particles and formation of dense clusters, while the case of consumption ($k < 0$) leads to dispersion of particles and formation of more uniform distributions. The latter case is more likely to exhibit hyperuniformity.



(a) Production $[\sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t))]$



(b) Consumption $[-\sum_{j=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t))]$

Figure 1: Snapshots of particle configurations for (a) production and (b) consumption of chemoattractant. The parameters are $\alpha = 0.5$, $D = 0.001$, $D_c = 0.1$, $\mu = -0.001$, and $N = 2000$ particles in an $L = 20$ domain. The positions and velocity direction of the particles are shown as arrows.

1.2 Continuum model

It is convenient to describe collective behavior using a particle density field $\rho(\mathbf{r}, t) = \sum_{i=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i(t))$ (Note that the $\rho(\mathbf{r}, t)$ here is particle number density, not probability density). By applying the Dean's method [1], we can derive the following stochastic partial differential equations for ρ and c:

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\alpha \nabla \cdot (\rho \nabla c) + D \nabla^2 \rho - \nabla \cdot \left[\sqrt{2D\rho}\boldsymbol{\Lambda}\right] \ , \tag{4a}$$

$$\frac{\partial}{\partial t} c(\mathbf{r}, t) = D_c \nabla^2 c + \mu c - \rho \ , \tag{4b}$$

where $\boldsymbol{\Lambda}(\mathbf{r}, t)$ is a Gaussian white noise with zero mean and unit variance, satisfying

$$\langle \Lambda_a(\mathbf{r}, t) \Lambda_b(\mathbf{r}', t') \rangle = \delta_{ab} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') \ . \tag{5}$$

Under the mean-field approximation, $\langle \boldsymbol{\Lambda} \rangle = 0$ leads to $\left\langle -\nabla \cdot \left[\sqrt{2D\rho}\boldsymbol{\Lambda}\right] \right\rangle \approx 0$, and we can neglect the noise term. This gives us the classical Keller-Segel model [2, 3]:

$$\frac{\partial}{\partial t}\rho\left(\mathbf{r},t\right)=-\alpha\nabla\cdot\left(\rho\nabla c\right)+D\nabla^{2}\rho\ , \tag{6a}$$

$$\frac{\partial}{\partial t}c\left(\mathbf{r},t\right)=D_{c}\nabla^{2}c+\mu c-\rho\ . \tag{6b}$$

One can perform linear stability analysis on the Keller-Segel model to determine the conditions for pattern formation. However, the mean-field approximation neglects fluctuations, which can be crucial for understanding the emergence of hyperuniformity. Therefore, we need to consider the full stochastic equations with noise to capture the effects of fluctuations on the system's behavior.

One obvious solution of Eqs. (6) is the homogeneous state $\rho(\mathbf{r},t)=\rho_0$ and $c(\mathbf{r},t)=c_0$, where $c_0=\rho_0/\mu$, which can be regarded as solution to Eqs. (4) in the mean-field sense. To analyze the stability of this homogeneous state, we can introduce small perturbations $\delta\rho(\mathbf{r},t)$ and $\delta c(\mathbf{r},t)$ around the homogeneous solution:

$$\rho(\mathbf{r},t)=\rho_0+\delta\rho(\mathbf{r},t),\quad c(\mathbf{r},t)=c_0+\delta c(\mathbf{r},t)\ . \tag{7}$$

Introducing these perturbations into Eqs. (4) gives us the following linearized equations:

$$\frac{\partial}{\partial t}\delta\rho=-\alpha\rho_0\nabla^2\delta c+D\nabla^2\delta\rho-\nabla\cdot\left[\sqrt{2D\rho_0}\mathbf{\Lambda}\right]\ , \tag{8a}$$

$$\frac{\partial}{\partial t}\delta c=D_c\nabla^2\delta c+\mu\delta c-\delta\rho\ . \tag{8b}$$

By performing Fourier transform on the linearized equations, we can analyze the growth rates of different modes and determine the conditions for instability. The presence of noise can lead to fluctuations that may drive the system towards hyperuniform states, even in regimes where the mean-field analysis predicts stability. Therefore, it is essential to consider the full stochastic dynamics to understand the emergence of hyperuniformity in chemotactic active matter.

We take the Fourier expansion under periodic boundary conditions:

$$\delta\rho(\mathbf{r},t)=\sum_{\mathbf{k}}\delta\rho_{\mathbf{k}}(t)e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}},\quad \delta c(\mathbf{r},t)=\sum_{\mathbf{k}}\delta c_{\mathbf{k}}(t)e^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\ . \tag{9}$$

Equivalently, it can also be written as a continuous Fourier transform:

$$\delta\rho_{\mathbf{k}}(t)=\int\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\delta\rho\left(\mathbf{r},t\right)\mathrm{d}\mathbf{k},\quad \delta c_{\mathbf{k}}(t)=\int\mathrm{e}^{-\mathrm{i}\mathbf{k}\cdot\mathbf{r}}\delta c\left(\mathbf{r},t\right)\mathrm{d}\mathbf{k}\ . \tag{10}$$

The specific normalization factor does not affect the symbol structure here. There is a subtlety in the Fourier transform of the noise term, which can be expressed as:

$$\nabla\cdot\left[\sqrt{2D\rho_0}\mathbf{\Lambda}\right]=\sum_{\mathbf{k}}\mathrm{i}\mathbf{k}\cdot\sqrt{2D\rho_0}\mathbf{\Lambda}_{\mathbf{k}}\ . \tag{11}$$

After Fourier transform, the linearized equations become:

$$\frac{\partial}{\partial t}\delta\rho_{\mathbf{k}}=-Dk^2\delta\rho_{\mathbf{k}}+\alpha\rho_0k^2\delta c_{\mathbf{k}}+\eta_{\mathbf{k}}\ , \tag{12a}$$

$$\frac{\partial}{\partial t}\delta c_{\mathbf{k}}=-(D_ck^2-\mu)\delta c_{\mathbf{k}}-\delta\rho_{\mathbf{k}}\ , \tag{12b}$$

where $\eta_{\mathbf{k}}=-\mathrm{i}\mathbf{k}\cdot\sqrt{2D\rho_0}\mathbf{\Lambda}_{\mathbf{k}}$. The noise term $\eta_{\mathbf{k}}$ has zero mean and its correlation can be calculated as follows:

$$\langle\eta_{\mathbf{k}}(t)\eta_{\mathbf{k}'}(t')\rangle=2D\rho_0k^2\delta(t-t')\ . \tag{13}$$

Eqs. (12) can be rewritten in matrix form as

$$\partial_t\begin{bmatrix}\delta\rho_{\mathbf{k}}\\ \delta c_{\mathbf{k}}\end{bmatrix}=A\left(k\right)\begin{bmatrix}\delta\rho_{\mathbf{k}}\\ \delta c_{\mathbf{k}}\end{bmatrix}+\begin{bmatrix}\eta_{\mathbf{k}}\\ 0\end{bmatrix}\ , \tag{14}$$

where

$$A(k) = \begin{bmatrix} -Dk^2 & \alpha\rho_0 k^2 \\ -1 & \mu - D_c k^2 \end{bmatrix} . \tag{15}$$

Let s be the linear growth rate, which is the eigenvalue of $A(k)$. The characteristic equation is given by

$$s^2 - (\mathrm{Tr}A)s + \det A = 0 . \tag{16}$$

where $\mathrm{Tr}A = -Dk^2 + \mu - D_c k^2$ and $\det A = Dk^2(D_c k^2 - \mu) + \alpha\rho_0 k^2$. The homogeneous state is linearly unstable if there exists a wavevector k such that the real part of s is positive. This leads to the stability condition:

$$\mathrm{Tr}A < 0, \quad \det A > 0 , \tag{17}$$

which leads to the following inequalities:

$$\begin{cases} \mu < (D + D_c)k^2 , \\ \mu < D_c k^2 + \frac{\alpha\rho_0}{D} . \end{cases} \tag{18}$$

Therefore, the homogeneous state is linearly stable if $\mu < \min\{(D + D_c)k^2, D_c k^2 + \frac{\alpha\rho_0}{D}\}$ for all k. In particular, for small k, the stability condition reduces to $\mu < 0 < \frac{\alpha\rho_0}{D}$, which means that the homogeneous state is stable if the chemoattractant decays and the chemotactic sensitivity is positive

$$\mu < 0, \quad \alpha > 0 . \tag{19}$$

However, even in the stable regime, fluctuations can still play a significant role in shaping the spatial distribution of particles, which is crucial for understanding the emergence of hyperuniformity.

1.2.1 Structure factor

In the limit of fast chemical diffusion, we can assume that the chemical field c quickly reaches a quasi-steady state, which allows us to eliminate c from the equations. Setting $\frac{\partial}{\partial t}\delta c_{\mathbf{k}} \approx 0$ in Eq. (12) gives us:

$$\delta c_{\mathbf{k}} \approx -\frac{\delta\rho_{\mathbf{k}}}{D_c k^2 - \mu} . \tag{20}$$

Substituting this back into the equation for $\delta\rho_{\mathbf{k}}$ yields:

$$\partial_t \delta\rho_{\mathbf{k}} = -k^2 \left[D + \frac{\alpha\rho_0}{D_c k^2 - \mu} \right] \delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}} . \tag{21}$$

Define effective diffusion coefficient $D_{\text{eff}}(k) = D + \frac{\alpha\rho_0}{D_c k^2 - \mu}$, we can rewrite the equation as:

$$\partial_t \delta\rho_{\mathbf{k}} = -k^2 D_{\text{eff}}(k)\delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}} . \tag{22}$$

This equation is an Ornstein-Uhlenbeck process, and we can calculate the steady-state structure factor, which is the variance of density fluctuations at wavevector $\mathbf{k}$, using the formula for the variance of an Ornstein-Uhlenbeck process (see Appendix 2.1):

$$S(k) = \langle \delta\rho_{\mathbf{k}} \delta\rho_{-\mathbf{k}} \rangle = \frac{2D\rho_0 k^2}{2k^2 D_{\text{eff}}(k)} = \frac{D\rho_0}{D + \frac{\alpha\rho_0}{D_c k^2 - \mu}} . \tag{23}$$

Here, the structure factor $S(k)$ can be interpreted as a noise–relaxation balance. The numerator $2D\rho_0 k^2$ represents the strength of the noise driving the fluctuations, while the denominator $2k^2 D_{\text{eff}}(k)$ represents the relaxation rate of the fluctuations due to diffusion and chemotactic interactions.

In an infinitely large system, the behavior of $S(k)$ as $k \to 0$ is crucial for determining whether the system is hyperuniform. Hyperuniformity is characterized by the suppression of density fluctuations at large scales, which corresponds to $S(k) \to 0$ as $k \to 0$ [4]. From Eq. (23), we can analyze the limit

$$\lim_{k \to 0} S(k) = S_0 = \frac{D\rho_0}{D - \frac{\alpha\rho_0}{\mu}} . \tag{24}$$

For the stable regime where $\mu < 0$ and $\alpha > 0$, S_0 is lower than the value of $S(k)$ for a Poisson process (which is $S(k) = \rho_0$ for all k), indicating that the system exhibits suppressed density fluctuations at large scales. However, S_0 is not zero, which means that the system is not perfectly hyperuniform. Instead, it is a "hyperuniform-like" state where density fluctuations are reduced but not completely eliminated at large scales.

If we do not assume fast chemical diffusion, we can solve the full two-dimensional linear Ornstein-Uhlenbeck system given by Eq. (14) to obtain the structure factor $S(k)$. The detailed calculation is provided in Appendix 2.2, and the final expression for $S(k)$ is given by

$$S(k) = D\rho_0 k^2 \frac{Dk^2(D_c k^2 - \mu) + \alpha\rho_0 k^2 + (D_c k^2 - \mu)^2}{D^2 k^4 (D_c k^2 - \mu) + D\alpha\rho_0 k^4 + Dk^2(D_c k^2 - \mu)^2 + \alpha\rho_0 k^2 (D_c k^2 - \mu)} . \tag{25}$$

This expression appears complicated, but it can be simplified into the same form as Eq. (24) when $k \to 0$, which confirms the consistency between the complete linear result and the fast chemical field approximation in the long-wavelength limit. The full expression for $S(k)$ also allows us to analyze the behavior of density fluctuations at intermediate and large wavevectors, which can provide insights into the spatial structure of the system at different length scales.

1.2.2 Hyperuniformity

From the expression for $S(k)$, we can analyze the conditions under which the system exhibits hyperuniformity. In the stable regime where $\mu < 0$ and $\alpha > 0$, we have $S_0 > 0$, which means that the system is not perfectly hyperuniform. However, we can achieve a state that is close to hyperuniform by tuning the parameters to following three conditions:

- $D \to 0$, which means that the particles do not diffuse on their own and their movement is solely driven by chemotactic interactions. But the Dean-noise term also vanishes in this limit, and the linear steady-state fluctuations naturally tend to zero. This is a trivial way to achieve hyperuniformity, where deterministic exclusion gradually smooths out initial fluctuations without continuous noise injection.

- $\mu \to 0^-$ and the chemical field is in a quasi-static state. If μ is close to zero, the chemical field satisfies a Poisson-like equation
$$0 \approx D_c \nabla^2 c - \rho , \tag{26}$$
which means that the chemical field is directly determined by the particle density distribution. In this case,
$$\delta c_{\mathbf{k}} \sim -\frac{\delta\rho_{\mathbf{k}}}{D_c k^2} . \tag{27}$$
Introducing this into Eq. (12a) gives us
$$\partial_t \delta\rho_{\mathbf{k}} \sim -\frac{\alpha\rho_0}{D_c}\delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}} , \tag{28}$$
which is an Ornstein-Uhlenbeck process with a k-independent relaxation rate, with the structure factor
$$S(k) = \langle \delta\rho_{\mathbf{k}} \delta\rho_{-\mathbf{k}} \rangle = \frac{\Gamma_k}{\lambda_k} \sim k^2 , \tag{29}$$
which indicates that the system is hyperuniform with class I scaling.

  Physically, when μ is close to zero, the chemical field can be regarded as a long-range field that mediates effective long-range interactions between particles. This can lead to strong suppression of density fluctuations at large scales, resulting in hyperuniformity.

2 Appendix

2.1 Calculation of variance for Ornstein-Uhlenbeck process

For an Ornstein-Uhlenbeck process described by the equation

$$\partial_t x(t) = -\lambda x(t) + \eta(t) \ , \tag{30}$$

where $\eta(t)$ is a Gaussian white noise with zero mean and correlation $\langle \eta(t)\eta(t')\rangle = 2\Gamma\delta(t-t')$, the formal solution is

$$X\left(t\right) = X\left(0\right) \mathrm{e}^{-\lambda t} + \int_{-\infty}^{t} \mathrm{e}^{-\lambda(t-s)} \eta\left(s\right) \mathrm{d}s \ . \tag{31}$$

After a long time, the initial condition becomes irrelevant:

$$X\left(t\right) \approx \int_{-\infty}^{t} \mathrm{e}^{-\lambda(t-s)} \eta\left(s\right) \mathrm{d}s \ . \tag{32}$$

The variance of $X(t)$ can be calculated as:

$$\begin{aligned} \langle X^2(t)\rangle &= \int_{-\infty}^{t}\int_{-\infty}^{t} \mathrm{e}^{-\lambda(2t-s-s')}\langle \eta(s)\eta(s')\rangle \mathrm{d}s\mathrm{d}s' \\ &= 2\Gamma \int_{-\infty}^{t} e^{-2\lambda(t-s)} \mathrm{d}s \\ &= \frac{\Gamma}{\lambda} \ . \end{aligned} \tag{33}$$

2.2 Structure factor from Lyapunov equation

The complete expression for $S(k)$ can be obtained directly from the steady-state covariance equation of the two-dimensional linear Ornstein-Uhlenbeck system, i.e., the Lyapunov equation. We start from the Fourier-space linearized system in matrix form:

$$\frac{d}{dt}\begin{pmatrix}\delta\rho_{\mathbf{k}} \\ \delta c_{\mathbf{k}}\end{pmatrix} = A(k)\begin{pmatrix}\delta\rho_{\mathbf{k}} \\ \delta c_{\mathbf{k}}\end{pmatrix} + \begin{pmatrix}\eta_{\mathbf{k}} \\ 0\end{pmatrix} , \tag{34}$$

where

$$A(k) = \begin{pmatrix} -Dk^2 & \alpha\rho_0 k^2 \\ -1 & -(D_c k^2 - \mu)\end{pmatrix} . \tag{35}$$

To simplify the notation, we define

$$a = Dk^2, \quad b = \alpha\rho_0 k^2, \quad q = D_c k^2 - \mu, \quad \Gamma = D\rho_0 k^2, \tag{36}$$

so that

$$A = \begin{pmatrix} -a & b \\ -1 & -q \end{pmatrix} , \tag{37}$$

and the noise covariance matrix is

$$Q = \begin{pmatrix} 2\Gamma & 0 \\ 0 & 0 \end{pmatrix} . \tag{38}$$

Defining covariance matrix C in component form as

$$C = \begin{pmatrix} x & y \\ y & z \end{pmatrix} , \tag{39}$$

where $x = \langle \delta\rho_{\mathbf{k}}\delta\rho_{-\mathbf{k}}\rangle = S(k)$, $y = \langle \delta\rho_{\mathbf{k}}\delta c_{-\mathbf{k}}\rangle$, $z = \langle \delta c_{\mathbf{k}}\delta c_{-\mathbf{k}}\rangle$. The Lyapunov equation for the steady-state covariance is given by

$$AC + CA^T + Q = 0 \ , \tag{40}$$

which can be explicitly written as three independent algebraic equations:

$$ax - by = \Gamma \, , \tag{41a}$$
$$x + (a+q)y - bz = 0 \, , \tag{41b}$$
$$y = -qz \, . \tag{41c}$$

From Eq. (41c) we have $z = -y/q$. Substituting into Eq. (41b) gives

$$y = -\frac{xq}{q(a+q)+b}. \tag{42}$$

Inserting this into Eq. (41a) and solving for x, we obtain

$$x = \Gamma \, \frac{q(a+q)+b}{(a+q)(aq+b)}. \tag{43}$$

Since $x = S(k)$, the structure factor reads

$$S(k) = \Gamma \, \frac{q(a+q)+b}{(a+q)(aq+b)} \, . \tag{44}$$

Substituting back the original variables a, b, q, and Γ, we arrive at the final expression:

$$S(k) = D\rho_0 k^2 \, \frac{\alpha\rho_0 k^2 + (D_c k^2 - \mu)\left[(D+D_c)k^2 - \mu\right]}{\left[(D+D_c)k^2 - \mu\right]\left[Dk^2(D_c k^2 - \mu) + \alpha\rho_0 k^2\right]} \, . \tag{45}$$

The denominator and numerator of Eq. (45) can be expanded as

$$\text{Denominator: } D^2k^4(D_c k^2 - \mu) + D\alpha\rho_0 k^4 + Dk^2(D_c k^2 - \mu)^2 + \alpha\rho_0 k^2(D_c k^2 - \mu) \, , \tag{46}$$
$$\text{Numerator: } Dk^2(D_c k^2 - \mu) + (D_c k^2 - \mu)^2 + \alpha\rho_0 k^2 \, . \tag{47}$$

Equation (45) is valid only when the linear system is stable, i.e., both eigenvalues of $A(k)$ have negative real parts. The stability conditions are

$$a + q > 0, \quad aq + b > 0, \tag{48}$$

which, in terms of the original parameters, read

$$(D+D_c)k^2 - \mu > 0, \quad Dk^2(D_c k^2 - \mu) + \alpha\rho_0 k^2 > 0 \, . \tag{49}$$

Equivalently,

$$\mu < (D+D_c)k^2, \quad \alpha\rho_0 + D(D_c k^2 - \mu) > 0 \, . \tag{50}$$

If these conditions are not satisfied, the Lyapunov equation does not admit a positive-definite steady-state solution, and the structure factor is not well-defined as a steady-state quantity.

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